Computation Flowchart – Controller

Reference Trajectory

$\boldsymbol{x}_{c,d},\ \dot{\boldsymbol{x}}_{c,d},\ \ddot{\boldsymbol{x}}_{c,d},\ \boldsymbol{x}_{c,d}^{(3)}$ $\boldsymbol{b}_{1,d},\ \dot{\boldsymbol{b}}_{1,d},\ \ddot{\boldsymbol{b}}_{1,d}$

$\boldsymbol{r}_{e,d},\ \dot{\boldsymbol{r}}_{e,d},\ \ddot{\boldsymbol{r}}_{e,d}$ $\boldsymbol{b}_{1,d,e},\ \dot{\boldsymbol{b}}_{1,d,e},\ \ddot{\boldsymbol{b}}_{1,d,e}$

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Control – Transformed State

$$\boldsymbol{X} = (\boldsymbol{r}_0, \boldsymbol{R}_0, \boldsymbol{q})$$

$$\boldsymbol{V} = \begin{bmatrix} \boldsymbol{v}_0{}^T & \boldsymbol{\omega}_0{}^T & \dot{\boldsymbol{q}}^T \end{bmatrix}^T$$

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{R}_0 & -\boldsymbol{R}_0 \hat{\boldsymbol{r}}_{0c}^0 & \boldsymbol{R}_0 \boldsymbol{A} \\ \boldsymbol{0} & \boldsymbol{I}_3 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{N}_1 & \boldsymbol{I}_n \end{bmatrix}$$

$$\boldsymbol{\xi} = \boldsymbol{T}\boldsymbol{V} \qquad \boldsymbol{\xi} = \begin{bmatrix} \dot{\boldsymbol{x}}_c \\ \boldsymbol{\omega}_0 \\ \boldsymbol{\rho} \end{bmatrix}$$

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Control – Disturbance Observer (GMO)

$$\boldsymbol{K}_o = \mathrm{diag}\left(\boldsymbol{K}_{o,t}, \boldsymbol{K}_{o,r}, \boldsymbol{K}_{o,\rho}\right) \succ 0, \qquad \boldsymbol{K}_{o,t}, \boldsymbol{K}_{o,r} \in \mathbb{R}^{3\times 3}, \quad \boldsymbol{K}_{o,\rho} \in \mathbb{R}^{n\times n}.$$

$$\hat{\boldsymbol{p}}(0) = \boldsymbol{p}(0) = \tilde{\boldsymbol{M}}(\boldsymbol{q}(0))\boldsymbol{\xi}(0)$$

$$\boldsymbol{p} = \tilde{\boldsymbol{M}}\boldsymbol{\xi}$$

$$\hat{\boldsymbol{d}}_e = \boldsymbol{K}_o\left(\boldsymbol{p} - \hat{\boldsymbol{p}}\right) \quad \Rightarrow \quad \dot{\hat{\boldsymbol{p}}} = \tilde{\boldsymbol{C}}^T\boldsymbol{\xi} - \tilde{\boldsymbol{g}} + \boldsymbol{u} + \hat{\boldsymbol{d}}_e$$

$$\hat{\boldsymbol{d}}_e = \begin{bmatrix} \hat{\boldsymbol{d}}_t^T & \hat{\boldsymbol{d}}_r^T & \hat{\boldsymbol{d}}_\rho^T \end{bmatrix}^T$$

Note that $\dot{\hat{\boldsymbol{p}}}$ will be updated after $\boldsymbol{u}$ is computed.

Computation Flowchart – Controller

Control – Translation

$$\boldsymbol{e}_x = \boldsymbol{x}_c - \boldsymbol{x}_{c,d}$$
$$\boldsymbol{e}_{v_x} = \dot{\boldsymbol{x}}_c - \dot{\boldsymbol{x}}_{c,d}$$

$$\boldsymbol{f}_d = -k_x \boldsymbol{e}_x - k_v \boldsymbol{e}_{v_x} + m\ddot{\boldsymbol{x}}_{c,d} + mg\boldsymbol{e}_3 - \hat{\boldsymbol{d}}_t$$

$$u_1 = \boldsymbol{f}_d^T \boldsymbol{R}_0 \boldsymbol{e}_3$$

$$\ddot{\boldsymbol{x}}_c = \frac{u_1}{m}\boldsymbol{R}_0\boldsymbol{e}_3 - g\boldsymbol{e}_3 + \frac{1}{m}\hat{\boldsymbol{d}}_t$$
$$\boldsymbol{e}_{a_x} = \ddot{\boldsymbol{x}}_c - \ddot{\boldsymbol{x}}_{c,d}$$

$$\dot{\boldsymbol{f}}_d = -k_x \boldsymbol{e}_{v_x} - k_v \boldsymbol{e}_{a_x} + m\boldsymbol{x}_{c,d}^{(3)}$$
$$\dot{u}_1 = \dot{\boldsymbol{f}}_d^T \boldsymbol{R}_0 \boldsymbol{e}_3 + \boldsymbol{f}_d^T \boldsymbol{R}_0 \hat{\boldsymbol{\omega}}_0 \boldsymbol{e}_3$$
$$\boldsymbol{x}_c^{(3)} = \frac{\dot{u}_1}{m}\boldsymbol{R}_0\boldsymbol{e}_3 + \frac{u_1}{m}\boldsymbol{R}_0\hat{\boldsymbol{\omega}}_0\boldsymbol{e}_3$$
$$\boldsymbol{e}_{j_x} = \boldsymbol{x}_c^{(3)} - \boldsymbol{x}_{c,d}^{(3)}$$

$$\ddot{\boldsymbol{f}}_d = -k_x \boldsymbol{e}_{a_x} - k_v \boldsymbol{e}_{j_x} + m\boldsymbol{x}_{c,d}^{(4)}$$

Computation Flowchart – Controller

Control – Rotation

$$\boldsymbol{b}_{3,c} = \frac{\boldsymbol{f}_d}{\|\boldsymbol{f}_d\|} \Rightarrow \boldsymbol{P}_1 = \boldsymbol{I}_3 - \boldsymbol{b}_{3,c}\boldsymbol{b}_{3,c}^T \Rightarrow \boldsymbol{s} = \boldsymbol{P}_1\boldsymbol{b}_{1,d} \Rightarrow \dot{\boldsymbol{b}}_{3,c} = \frac{1}{\|\boldsymbol{f}_d\|}\boldsymbol{P}_1\dot{\boldsymbol{f}}_d$$

$$\boldsymbol{b}_{1,c} = \frac{\boldsymbol{s}}{\|\boldsymbol{s}\|} \Rightarrow \begin{aligned} \dot{\boldsymbol{P}}_1 &= -\left(\dot{\boldsymbol{b}}_{3,c}\boldsymbol{b}_{3,c}^T + \boldsymbol{b}_{3,c}\dot{\boldsymbol{b}}_{3,c}^T\right) \\ \boldsymbol{P}_2 &= \boldsymbol{I}_3 - \boldsymbol{b}_{1,c}\boldsymbol{b}_{1,c}^T \end{aligned} \Rightarrow \dot{\boldsymbol{s}} = \dot{\boldsymbol{P}}_1\boldsymbol{b}_{1,d} + \boldsymbol{P}_1\dot{\boldsymbol{b}}_{1,d} \Rightarrow \begin{aligned} \dot{\boldsymbol{b}}_{1,c} &= \frac{1}{\|\boldsymbol{s}\|}\boldsymbol{P}_2\dot{\boldsymbol{s}} \\ \dot{\boldsymbol{P}}_2 &= -\left(\dot{\boldsymbol{b}}_{1,c}\boldsymbol{b}_{1,c}^T + \boldsymbol{b}_{1,c}\dot{\boldsymbol{b}}_{1,c}^T\right) \end{aligned}$$

$$\Rightarrow \begin{aligned} \boldsymbol{b}_{2,c} &= \boldsymbol{b}_{3,c} \times \boldsymbol{b}_{1,c} \\ \boldsymbol{R}_{0,c} &= \begin{bmatrix} \boldsymbol{b}_{1,c} & \boldsymbol{b}_{2,c} & \boldsymbol{b}_{3,c} \end{bmatrix} \end{aligned} \Rightarrow \begin{aligned} \dot{\boldsymbol{b}}_{2,c} &= \dot{\boldsymbol{b}}_{3,c} \times \boldsymbol{b}_{1,c} + \boldsymbol{b}_{3,c} \times \dot{\boldsymbol{b}}_{1,c} \\ \dot{\boldsymbol{R}}_{0,c} &= \begin{bmatrix} \dot{\boldsymbol{b}}_{1,c} & \dot{\boldsymbol{b}}_{2,c} & \dot{\boldsymbol{b}}_{3,c} \end{bmatrix} \end{aligned}$$

$$\Rightarrow \boldsymbol{\omega}_{0,c} = \left(\boldsymbol{R}_{0,c}^T\dot{\boldsymbol{R}}_{0,c}\right)^\vee$$

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$$\ddot{\boldsymbol{b}}_{3,c} = \frac{1}{\|\boldsymbol{f}_d\|}\left(\dot{\boldsymbol{P}}_1\dot{\boldsymbol{f}}_d + \boldsymbol{P}_1\ddot{\boldsymbol{f}}_d\right) - \frac{\boldsymbol{b}_{3,c}^T\dot{\boldsymbol{f}}_d}{\|\boldsymbol{f}_d\|^2}\boldsymbol{P}_1\dot{\boldsymbol{f}}_d \quad \Rightarrow \quad \ddot{\boldsymbol{P}}_1 = -(\ddot{\boldsymbol{b}}_{3,c}\boldsymbol{b}_{3,c}^T + 2\dot{\boldsymbol{b}}_{3,c}\dot{\boldsymbol{b}}_{3,c}^T + \boldsymbol{b}_{3,c}\ddot{\boldsymbol{b}}_{3,c}^T)$$

$$\ddot{\boldsymbol{s}} = \ddot{\boldsymbol{P}}_1\boldsymbol{b}_{1,d} + 2\dot{\boldsymbol{P}}_1\dot{\boldsymbol{b}}_{1,d} + \boldsymbol{P}_1\ddot{\boldsymbol{b}}_{1,d}$$

$$\ddot{\boldsymbol{b}}_{1,c} = \frac{1}{\|\boldsymbol{s}\|}(\dot{\boldsymbol{P}}_2\dot{\boldsymbol{s}} + \boldsymbol{P}_2\ddot{\boldsymbol{s}}) - \frac{\boldsymbol{b}_{1,c}^T\dot{\boldsymbol{s}}}{\|\boldsymbol{s}\|}\dot{\boldsymbol{b}}_{1,c} \quad \Rightarrow \quad \ddot{\boldsymbol{b}}_{2,c} = \ddot{\boldsymbol{b}}_{3,c} \times \boldsymbol{b}_{1,c} + 2\dot{\boldsymbol{b}}_{3,c} \times \dot{\boldsymbol{b}}_{1,c} + \boldsymbol{b}_{3,c} \times \ddot{\boldsymbol{b}}_{1,c}$$

$$\ddot{\boldsymbol{R}}_{0,c} = \begin{bmatrix}\ddot{\boldsymbol{b}}_{1,c} & \ddot{\boldsymbol{b}}_{2,c} & \ddot{\boldsymbol{b}}_{3,c}\end{bmatrix}$$

$$\Rightarrow \quad \dot{\boldsymbol{\omega}}_{0,c} = \left(\dot{\boldsymbol{R}}_{0,c}^T\dot{\boldsymbol{R}}_{0,c} + \boldsymbol{R}_{0,c}^T\ddot{\boldsymbol{R}}_{0,c}\right)^{\vee}$$

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$$\boldsymbol{e}_R = \frac{1}{2}\left(\boldsymbol{R}_{0,c}^T\boldsymbol{R}_0 - \boldsymbol{R}_0^T\boldsymbol{R}_{0,c}\right)^\vee$$

$$\boldsymbol{e}_\omega = \boldsymbol{\omega}_0 - \boldsymbol{R}_0^T\boldsymbol{R}_{0,c}\boldsymbol{\omega}_{0,c}$$

$$\Rightarrow \quad \boldsymbol{u}_2 = \tilde{\boldsymbol{M}}_r\left[\boldsymbol{R}_0^T\boldsymbol{R}_{0,c}\dot{\boldsymbol{\omega}}_{0,c} - \hat{\boldsymbol{\omega}}_0\boldsymbol{R}_0^T\boldsymbol{R}_{0,c}\boldsymbol{\omega}_{0,c} - \tilde{\boldsymbol{M}}_{r,d}^{-1}\left(k_R\boldsymbol{e}_R + k_\omega\boldsymbol{e}_\omega\right)\right] + \tilde{\boldsymbol{C}}_r\boldsymbol{\omega}_0 + \tilde{\boldsymbol{C}}_{r\rho}\boldsymbol{\rho} - \hat{\boldsymbol{d}}_r$$

Computation Flowchart – Controller

Control – Arm

$$\boldsymbol{b}_{1,e} = \boldsymbol{R}_e \boldsymbol{e}_1$$
$$\boldsymbol{b}_{2,e} = \boldsymbol{R}_e \boldsymbol{e}_2$$
$$\boldsymbol{b}_{3,e} = \boldsymbol{R}_e \boldsymbol{e}_3$$

$$\boldsymbol{R}_e^0(\boldsymbol{q}) = \boldsymbol{R}_n^0(\boldsymbol{q}) \quad \boldsymbol{\omega}_e^0 = \boldsymbol{\omega}_n^0$$
$$\boldsymbol{\omega}_e = \left(\boldsymbol{R}_e^0\right)^T \boldsymbol{\omega}_e^0 = \boldsymbol{R}_0^e(\boldsymbol{\omega}_0 + \boldsymbol{J}_{\omega_e}^q \dot{\boldsymbol{q}})$$
$$\omega_{3,e} = \boldsymbol{\omega}_e \boldsymbol{e}_3$$

$\Rightarrow$

$$\boldsymbol{P}_{1,e} = \boldsymbol{I}_3 - \boldsymbol{b}_{3,e}\boldsymbol{b}_{3,e}^T$$
$$\dot{\boldsymbol{b}}_{3,e} = \boldsymbol{R}_e \hat{\boldsymbol{\omega}}_e \boldsymbol{e}_3$$

$\Rightarrow$

$$\boldsymbol{s}_e = \boldsymbol{P}_{1,e}\boldsymbol{b}_{1,e,d}$$

$$\boldsymbol{b}_{1,e,c} = \frac{\boldsymbol{s}_e}{\|\boldsymbol{s}_e\|}$$

$\Rightarrow$

$$\dot{\boldsymbol{P}}_{1,e} = -\left(\dot{\boldsymbol{b}}_{3,e}\boldsymbol{b}_{3,e}^T + \boldsymbol{b}_{3,e}\dot{\boldsymbol{b}}_{3,e}^T\right)$$
$$\boldsymbol{P}_{2,e} = I_3 - \boldsymbol{b}_{1,e,c}\boldsymbol{b}_{1,e,c}^T$$

$\Rightarrow$

$$\dot{\boldsymbol{s}}_e = \dot{\boldsymbol{P}}_{1,e}\boldsymbol{b}_{1,e,d} + \boldsymbol{P}_{1,e}\dot{\boldsymbol{b}}_{1,e,d}$$

$\Rightarrow$

$$\dot{\boldsymbol{b}}_{1,e,c} = \frac{1}{\|\boldsymbol{s}_e\|}\boldsymbol{P}_{2,e}\dot{\boldsymbol{s}}_e$$
$$\dot{\boldsymbol{P}}_{2,e} = -\left(\dot{\boldsymbol{b}}_{1,e,c}\boldsymbol{b}_{1,e,c}^T + \boldsymbol{b}_{1,e,c}\dot{\boldsymbol{b}}_{1,e,c}^T\right)$$

$\Rightarrow$

$$\omega_{3,e,c} = \boldsymbol{b}_{3,e}^T(\boldsymbol{b}_{1,e,c} \times \dot{\boldsymbol{b}}_{1,e,c})$$

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$$\dot{\boldsymbol{\omega}}_e = \boldsymbol{R}_0^e \left(\dot{\boldsymbol{\omega}}_0 + \boldsymbol{J}_{\omega_e}^q \ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}_{\omega_e}^q \dot{\boldsymbol{q}} + \hat{\boldsymbol{\omega}}_0 \boldsymbol{J}_{\omega_e}^q \dot{\boldsymbol{q}}\right)$$

$$\Rightarrow \quad \ddot{\boldsymbol{b}}_{3,e} = \boldsymbol{R}_e(\hat{\boldsymbol{\omega}}_e^2 + \hat{\dot{\boldsymbol{\omega}}}_e)\boldsymbol{e}_3$$

$$\Rightarrow \quad \ddot{\boldsymbol{P}}_{1,e} = -(\ddot{\boldsymbol{b}}_{3,e}\boldsymbol{b}_{3,e}^T + 2\dot{\boldsymbol{b}}_{3,e}\dot{\boldsymbol{b}}_{3,e}^T + \boldsymbol{b}_{3,e}\ddot{\boldsymbol{b}}_{3,e}^T)$$

$$\Rightarrow \quad \ddot{\boldsymbol{s}}_e = \ddot{\boldsymbol{P}}_{1,e}\boldsymbol{b}_{1,e,d} + 2\dot{\boldsymbol{P}}_{1,e}\dot{\boldsymbol{b}}_{1,e,d} + \boldsymbol{P}_{1,e}\ddot{\boldsymbol{b}}_{1,e,d}$$

$$\ddot{\boldsymbol{b}}_{1,e,c} = \frac{1}{\|\boldsymbol{s}_e\|}\left(\dot{\boldsymbol{P}}_{2,e}\dot{\boldsymbol{s}}_e + \boldsymbol{P}_{2,e}\ddot{\boldsymbol{s}}_e\right) - \frac{\boldsymbol{b}_{1,e,c}^T\dot{\boldsymbol{s}}_e}{\|\boldsymbol{s}_e\|}\dot{\boldsymbol{b}}_{1,e,c}$$

$$\Rightarrow \quad \dot{\omega}_{3,e,c} = \dot{\boldsymbol{b}}_{3,e}^T(\boldsymbol{b}_{1,e,c} \times \dot{\boldsymbol{b}}_{1,e,c}) + (\boldsymbol{b}_{3,e} \times \boldsymbol{b}_{1,e,c})^T\ddot{\boldsymbol{b}}_{1,e,c}$$

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$$\boldsymbol{e}_{x_E} = \boldsymbol{r}_e - \boldsymbol{r}_{e,d}$$
$$e_{R_{E,3}} = -\boldsymbol{b}_{1,e,c}^T \boldsymbol{b}_{2,e}$$

$$\Rightarrow \quad \boldsymbol{e}_y = \begin{bmatrix} \boldsymbol{e}_{x_E} \\ e_{R_{E,3}} \end{bmatrix}$$

$$\boldsymbol{e}_{v_E} = \dot{\boldsymbol{r}}_e - \dot{\boldsymbol{r}}_{e,d}$$
$$e_{\omega_{E,3}} = \omega_{3,e} - \omega_{3,e,c}$$

$$\Rightarrow \quad \boldsymbol{e}_{v_y} = \begin{bmatrix} \boldsymbol{e}_{v_E} \\ e_{\omega_{E,3}} \end{bmatrix}$$

$$\ddot{\boldsymbol{y}}_d = \begin{bmatrix} \ddot{\boldsymbol{r}}_{e,d} \\ \dot{\omega}_{3,e,c} \end{bmatrix}$$

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$$\boldsymbol{M}_y = \boldsymbol{M}_y^T \succ 0, \quad \boldsymbol{K}_y = \boldsymbol{K}_y^T \succ 0, \quad \boldsymbol{D}_y = \boldsymbol{D}_y^T \succ 0.$$

The translational block of $\boldsymbol{My}$ is isotropic.

$$\hat{\boldsymbol{F}}_y = \left(\boldsymbol{J}_y^{\dagger}\right)^T \hat{\boldsymbol{d}}_e = \bar{\boldsymbol{J}}_{1,y}\hat{\boldsymbol{d}}_t + \bar{\boldsymbol{J}}_{2,y}\hat{\boldsymbol{d}}_r + \bar{\boldsymbol{J}}_{3,y}\hat{\boldsymbol{d}}_\rho \qquad \boldsymbol{u}_1 = u_1\boldsymbol{R}_0\boldsymbol{e}_3$$

$$\begin{aligned}\boldsymbol{u}_3 = -\bar{\boldsymbol{J}}_{3,y}^{-1}\Bigg(&\begin{bmatrix}\bar{\boldsymbol{J}}_{1,y} & \bar{\boldsymbol{J}}_{2,y}\end{bmatrix}\begin{bmatrix}\boldsymbol{u}_1 - \tilde{\boldsymbol{g}}_t \\ \boldsymbol{u}_2 - \tilde{\boldsymbol{C}}_r\boldsymbol{\omega}_0 - \tilde{\boldsymbol{C}}_{r\rho}\boldsymbol{\rho}\end{bmatrix} + \boldsymbol{\Lambda}_y(\dot{\boldsymbol{J}}_y\boldsymbol{\xi} - \ddot{\boldsymbol{y}}_d) + \boldsymbol{\Lambda}_y\boldsymbol{M}_y^{-1}(\boldsymbol{D}_y\boldsymbol{e}_{v_y} + \boldsymbol{K}_y\boldsymbol{e}_y) \\ &-(\boldsymbol{\Lambda}_y\boldsymbol{M}_y^{-1} - \boldsymbol{I}_4)\hat{\boldsymbol{F}}_y\Bigg) - \tilde{\boldsymbol{C}}_{r\rho}^T\boldsymbol{\omega}_0 + \tilde{\boldsymbol{C}}_\rho\boldsymbol{\rho}\end{aligned}$$

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